NACA 2415 airfoil wind tunnel

EMAE 285 · Lab 6

Experiment overview

Measured surface pressure distributions on a NACA 2415 airfoil at 20 m/s over angles of attack from –10° to +20°. Investigated lift trends, stall and flow separation.

Variables and instrumentation

Pressure taps, airspeed, angle of attack

Analysis and presentation

Pressure coefficient distributions; lift trends; stall discussion

Engineering relevance

Shows experimental design and interpretation of fluid-mechanics data.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1STcNEN-m5aRaWFoqdU7kz8zvdi2gtzEZTOwQcJPuFwo/edit